1. **Domain and system**
   Healthcare: a patient and triage queue system

2. **System name**
   CareFlow

3. **Primary Users**
   Patients: need to submit intake information and receive care
   Nurses: need to assess incoming patients and set/adjust triage priority.
   Clinicians: need to see patients in priority order and record outcomes of evaluation
   Clinic Administrator: manage system

4. **Business Goals**
   CareFlow aims to improve the patient intake and triage process by helping staff identify urgent cases quickly, reduce unnecessary wait times, and keep patients informed about their status. Success means patients are prioritized consistently and safely, staff can manage the queue with less manual coordination, and all significant changes to a patient's priority or status are accurately recorded.

5. **Candidate MVP Features**
   1. A patient can complete and submit an intake form before joining the triage queue.
   2. A patient can check their current queue status and estimated wait information.
   3. A nurse can review a patient's intake information and assign a triage priority.
   4. A nurse can change a patient's queue priority when their condition or circumstances change.
   5. A clinician can view a patient's waiting for evaluation in priority order.
   6. A clinician can update a patient's status after evaluation.

Key Constraints:

- **Patient data sensitivity**: only authorized roles may access appropriate patient information.
- **Audit trail:** changes to triage priority and patient status should be recorded.
- **Role authorization:** patients, nurses, and clinicians must have distinct, enforced permissions.
- **Data integrity**: required intake information must be validated before the patient enters the queue.
- **Emergency/priority handling:** higher-priority patients must be able to move ahead of lower-priority patients according to clearly defined rules.

AI-Use Note:

We expect to use AI tools to brainstorm and refine requirements, review design decisions, assist with code and documentation, and generate test cases. We will verify AI-generated output carefully and avoid relying on AI-generated healthcare, privacy, or security logic without human review.